Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Antibody-Drug Conjugates for Advanced Prostate Cancer Therapy			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (v), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Bio Medical		Bio Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Therapeutic Antibodies	Medical Bio	Pharmaceutical Bio-based Technologies and systems
Concept of Proposed Technology	The proposed technology focuses on developing advanced treatments for metastatic castration-resistant prostate cancer (mCRPC) using the following technology: **Targeted ADC Production-** Developing a Prostate-Specific Membrane Antigen (PSMA)-targeted Antibody-Drug Conjugate (ADC) that links highly specific antibodies to potent cytotoxic payloads (such as modified docetaxel) using AI-optimized linkers. **Prostate Cancer Organoid Production -** Created through Source Cell Preparation, 3D Culture & Embedding, Organoid Induction & Growth, Maturation & Expansion using prostate cancer cell lines.			
Rationale for Joint R&D Project	**Technical Synergy:** While the industrial partner leads the ADC development and AI-based modifications, the specific expertise required for perfecting prostate cancer organoid models is held by the **Overseas Research Institute**.			

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	Innovation Gap: Joint research is necessary to establish currently unestablished prostate cancer organoid technologies, which will allow for more accurate predictions of human efficacy and a reduction in the loss of animal life. **Global Competitiveness:** Combining international research excellence with domestic industrial engineering maximizes the potential for successful global commercialization and patent acquisition.
Key Words	1. Prostate cancer treatment 2. Antibody-Drug-Conjugate (ADC) 3. Cancer organoid model

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2)]	**Start of the Project**	Level (3)	**End of the Project**	Level (5)
R&D Objectives	**Side Effect Reduction:** Developing a targeted ADC to significantly reduce the dosage and systemic toxicity of conventional treatments like docetaxel. **Methodological Innovation:** Completing a prostate cancer organoid system to replace animal testing in the drug validation process.			

R&D Target Specifications

Category	**Performance Metric**	**Target Specification**
Therapeutic Efficacy	Tumor growth inhibition	≥ 80% compared to untreated controls
Model Accuracy	Organoid experiment error rate	Within ±10% compared to traditional animal models
Antibody Suitability	Target specificity	Confirmed suitability through PSMA library screening
Process Timeline	Single-domain antibody completion	Scheduled for first half of 2026
Conjugation	Linker-Payload synthesis	Scheduled for 2026 (first/second half depending on candidate)

R&D Plan

ADC Development:

1. **Antibody Production:** Completing human and single-domain antibodies using cartilaginous fish immunization and library screening (Expected 1H 2026).
2. **Payload & Linker:** Using AI to modify docetaxel and other toxicants for optimal conjugation; exploring antibody-linking sites for cleavable linker production (Expected 2H 2026).

Organoid Development: Establishing 3D cultures using the 22Rv1 prostate cancer cell line to create reliable organoid models.

Validation: Conducting tumor growth inhibition experiments using the newly established organoids to confirm drug efficacy.

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**Scientific Validation:** Leading the development of prostate cancer organoids through specialized 3D culture techniques. **Technical Benchmarking:** Providing the foundational biological models (22Rv1 cell line) necessary for non-animal testing. **Collaborative Synergy:** Hosting dispatched personnel from the Korean partner to facilitate knowledge transfer and joint research.
Expected R&R of Korean SME Partner	**Core Technical Contribution:** Conducting antibody-linker and toxic drug-linker conjugation using proprietary AI prediction tools. **Industrial Development:** Leading the chemical modification of payloads (docetaxel) and the production of prostate-specific antibodies. **Testing & Validation:** Receiving organoid models from the research institute to perform final tumor growth inhibition assays. **Resource Investment:** Providing specialized human resources and capital to maximize mutual synergy.

5. Expected Outcomes/Impact

Expected Outcomes and Impact	**Technical & Academic:** Generation of international patent applications and high-impact scientific papers. **Economic Impact:** Technology transfer to global pharmaceutical companies, benefiting from a prostate cancer therapeutics market expected to reach nearly $22 billion by 2029. **Social & Ethical Impact:** Reducing the "unnecessary loss of life" by replacing animal testing with human-relevant organoid models. **Broad Application:** Successful development of the ADC platform for prostate cancer can potentially be applied to various other cancer types in the future.

6. References

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